

method, in cells A15:B19, results are arranged as follows (SS stands for Sum of Squares):

Resource A Linear and quadratic calibration (Excel).

	A	B	C	D	E	F
1	Resource A : Linear and quadratic calibration using OLS					
2	Theophylline Series 1 (µg/l)					
3	Linear model			Quadratic model		
4	X	Y (AU)		X	X ²	Y(AU)
5	0.02	0.293		0.02	0.0004	0.293
6	0.02	0.443		0.02	0.0004	0.443
7	0.10	1.874		0.10	0.0100	1.874
8	0.10	1.810		0.10	0.0100	1.810
9	0.50	8.904		0.50	0.2500	8.904
10	0.50	8.306		0.50	0.2500	8.306
11	2.50	23.411		2.50	6.2500	23.411
12	2.50	37.832		2.50	6.2500	37.832
13	10.00	124.835		10.00	100.0000	124.835
14	10.00	129.605		10.00	100.0000	129.605
15						
16	12.634	0.580	=LINEST(B5:B14;A5:A14;TRUE;TRUE)	0.084	11.779	1.040
17	0.332	1.530		0.188	1.949	1.915
18	0.995	3.9806		0.995	4.1963	#N/A
19	1452.238	8		653.499	7	#N/A
20	23011.141	126.762		23014.642	123.2614	#N/A
21						
22	Coefficient a0	0.580		Coefficient a0	1.040	
23	Coefficient a1	12.634		Coefficient a1	11.779	
24	Residual std dev	3.98061		Coefficient a2	0.084	
25	sa0	1.530		Residual std dev	4.1963	
26	sa1	0.332		sa0	1.915	
27	r ²	0.9945		sa1	1.949	
28				sa2	0.188	
29				r ²	0.9947	

When looking at the returned parameters, the straight-line model seems satisfactory, while the coefficient of determination r^2 is extremely high. This parameter, expressed as a percentage, can be interpreted as the percent of the variation of Y explained by the regression model (see Section 2.4.2). The difference with a perfect model corresponds to the amount of unexplained variation. In the example, about 99.5% of Y can be explained by the selected regression model, and 0.5% is unexplained. The opposite would have been surprising because the analytical method was purposely developed on this physicochemical property.

In the second line of the result matrix, the standard deviations of coefficients are also returned and can be used for further calculations. More details about r^2 are available in Section 2.4.2.

On the same worksheet, a second-order polynomial model is adjusted to the same data. Equation (2.25) specifies the quadratic model used in this example and the corresponding inverse calibration function. The three coefficients are also estimated by using built-in function LINEST with the arguments =LINEST (F5:F14;D5:E14;TRUE;TRUE) .

Quadratic calibration model

$$Y_i = a_0 + a_1X_i + a_2X_i^2 + E_i \quad (2.25)$$

Inverse quadratic calibration model

$$Z_k = \frac{-a_1 + \sqrt{a_1^2 - 4a_2(a_0 - Y_k)}}{2a_2} \quad (2.26)$$

It is well-known that the second-order equation has two workable solutions, similarly, the inverse calibration quadratic model may come down to two values. The solution presented in Eq. (2.26) assumes that the calibration curve is concave. This is the usual situation when Y is regularly increasing when X is increasing. Hence, to apply `LINEST` to this model with three parameters, a new column containing the squared values of X , called “ X^2 ” must first be added. The built-in function `LINEST` is very flexible and can also be used for many applications. For instance, it can cope with multiple linear regression models such as employed for the isotopic pattern deconvolution method described in Section 1.5.

It appears in the range E5 : E14. In cell D16, the formula which generates the results is typed. It is not displayed on the worksheet for lack of space, but it is easy to see that both columns D and E containing X and X^2 , respectively, are input as the second argument. The coefficient of determination r^2 of the quadratic model equals 0.995 and is identical to the linear model. Therefore, this criterion is not sufficient to select the best calibration model, and another approach is needed, as explained in many statistical handbooks [2].

To attempt to define a procedure suitable for the selection of the best model that can be estimated from the calibration data of Table 2.1 collected on one single day, replicate measurements were done on six samples spiked with known amounts of theophylline. Instrumental responses expressed in arbitrary units (AU) are reported in Table 2.2. No measurement is out of the calibration range as the lowest concentration here is 0.05 $\mu\text{g/l}$, and the lowest calibrator was 0.02 $\mu\text{g/l}$.

Table 2.2 THEOPHYLLINE series 1 – validation data.

Spike ($\mu\text{g/l}$)	Response (AU)	Inverse-predicted concentrations ($\mu\text{g/l}$)			
		Linear	Bias (%)	Quadratic	Bias (%)
0.05	1.307	0.058	15.1	0.023	−55
0.05	1.259	0.054	7.5	0.019	−63
0.10	1.909	0.105	5.2	0.074	−26
0.10	1.883	0.103	3.1	0.072	−28
0.50	8.638	0.638	27.6	0.642	28
0.50	8.786	0.650	29.9	0.655	31
1.00	18.154	1.391	39.1	1.438	44
1.00	17.672	1.353	35.3	1.398	40
2.50	39.004	3.041	21.7	3.152	26
2.50	37.949	2.958	18.3	3.067	23
10.00	123.565	9.735	−2.7	9.729	−3
10.00	126.487	9.966	−0.3	9.947	−1

Instrumental responses and inverse-predicted concentrations for six spiked samples.

This comment is made because it is always dangerous to extrapolate. With these new responses, inverse-predicted concentrations were calculated with both inverse models using the estimated model coefficients reported in Resource A and put together in the same table. It is additionally possible to compute the individual bias for each result as described in Section 4.1.1. Obtained bias values are hugely varying and disappointing, while both models seem remarkably similar when looking at Figure 2.1. These results are very unsatisfactory.

The reason for this behavior is that the calibration data do not respect one of the mandatory assumptions for applying the OLS regression: the variance of response Y must be constant over all the calibration range, and it is not the case here. Another algorithm must be applied, known as Weighted Least-Squares or WLS.

2.3.2 Weighted Least-squares (WLS) Regression

As explained, the implicit assumption for applying OLS method is that all random variables E_i have the same theoretical variances, denoted σ^2 . As explained before, the best estimate of this variance is the residual variance denoted s_E^2 and given by Eq. (2.15). This property, called homoscedasticity, means that the variances of the Y -values are homogeneous over all the calibration range. When the calibration range is large, it is not unusual that repeated instrumental responses be more spread out at high concentrations than at low concentrations.

Calibrator measurement dispersion often increases when the concentration increases and response variances are no longer identical throughout the calibration domain. When dealing with MU the relationship between concentration and response dispersion is a key issue. For instance, most LC-coupled methods (e.g., inductively coupled plasma-mass spectrometer [ICP-MS], liquid chromatography-mass spectrometer [LC-MS], etc.) cover a large calibration range, and Y -value variances significantly increase with analyte concentration. Because low concentrations have smaller variances, precision at the lower end of the range may therefore be compromised and impair the limits of quantification (LOQ).

The practical answer of statisticians was to develop the theory of WLS regression. The basic idea is to compensate for the variation in dispersion of the response Y , by weighting the residuals in the opposite manner to the increase in variability. The most logical weighting is to use the inverse of local variances of Y for each calibrator:

$$W_i = \frac{1}{s_{Y_i}^2}$$

This means it is necessary to make replicate measurements to have pertinent estimates of the variances. This may become rather cumbersome. In the context of calibration, simpler weighting schemes can be implemented to correct this heteroscedasticity issue and ensure a reliable estimation of the regression parameters. More empirical weighting factors are based either on the concentration of the calibrant or the measured response, providing a simpler means of approximation. In this case, several weights W_i are applicable, such as:

$$W_i = \frac{1}{X^2} \text{ or } \frac{1}{\sqrt{X}} \text{ or } \frac{1}{Y^2} \text{ or } \frac{1}{\sqrt{Y}}, \text{ etc.}$$

It is not always easy to select the best-adapted weights W_i but a practical approach is possible by making several trials. To present some details of the method, we have chosen a classic approach by using the inverse of the squared concentration $1/X^2$. A major criticism can be made of this choice:

- When the calibration range is large, the ratio between the weights applied to the smallest calibrator and the highest is enormous. In the example of Table 2.2, the smallest weight is $1/0.02^2 = 2500$, and the highest $1/10^2 = 0.01$. This 25,000-fold ratio may emphasize the importance of smallest concentration at the expense of the highest, which may become meaningless. It can also be assumed it may be of some consequence on the accuracy of inverse-predicted values.

Analysts must be aware of this possible issue because this weighting is often chosen and implicitly implemented in instrument monitoring software. Reasons are easy to understand. Whatever the calibration experimental design, without replicates, the computation is straightforward. The following example is given to illustrate the efficiency of the WLS despite this questionable weighting choice. Unfortunately, the built-in function `LINEST` does not present the weighted variant of the LS method. It is possible, with some programming effort, to build template sheets that provide the expected results, as illustrated in a publication [3]. The following formulas can be used to develop such a template, but it is limited to the straight-line model.

Classic weights

$$W_i = \frac{1}{X_i^2} \quad (2.27)$$

Weighted average of X

$$\bar{X} = \frac{\sum_i W_i X_i}{\sum_i W_i} \quad (2.28)$$

Weighted average of Y

$$\bar{Y} = \frac{\sum_i W_i Y_i}{\sum_i W_i} \quad (2.29)$$

Sum of squares for X

$$SS(X) = \sum_i W_i X_i^2 - \frac{(\sum_i W_i X_i)^2}{\sum_i W_i} \quad (2.30)$$

Sum of squares for Y

$$SS(Y) = \sum_i W_i Y_i^2 - \frac{(\sum_i W_i Y_i)^2}{\sum_i W_i} \quad (2.31)$$

Sum of cross-products

$$SP(X, Y) = \sum_i W_i X_i Y_i - \frac{\sum_i W_i X_i \sum_i W_i Y_i}{\sum_i W_i} \quad (2.32)$$

Slope or sensitivity

$$a_1 = \frac{SP(X, Y)}{SS(X)} \quad (2.13)$$

Intercept or blank

$$a_0 = \bar{Y} - a_1 \bar{X} \quad (2.14)$$

Another opportunity that we recommend to apply WLS to any model is to program a short Python script, such as Resource B. Likewise, in other Python programs, comments are added in plain text to explain how it works.

Resource B Calibration using OLS and WLS (Python).

It is necessary to import the complementary packages, mainly `statsmodels.formula.api`, that contains all linear regression functions.

```
import pandas as pd
import statsmodels.formula.api as sm
```

For this example, the THEOPHYLLINE series 1 data are used. If a more general program must be developed, Python allows many possibilities to read external data in many formats.

```
X = [0.02, 0.02, 0.10, 0.10, 0.50, 0.50, 2.50, 2.50, 10.00,
      10.00]
Y = [0.293, 0.443, 1.874, 1.810, 8.904, 8.306, 23.411, 37.832,
      124.835, 129.605]
```

Complementary variables are added for the quadratic model or the weighting of data.

```
X2 = [i*i for i in X] # squares of X
```

Select the weighting. This underlines the simplicity of this choice, while using another weight may be more cumbersome.

```
W = [1/i**2 for i in X] # weights
```

All variables are organized in a data frame. This step is necessary to apply the regression function of Python.

```
df_data = pd.DataFrame({"x":X, "x2": X2, "y":Y})
```

The calibration function is declared in a straightforward way, as a text using the variable text names: `formula = 'y ~ x'`. Thereafter, it is combined with the data frame into a model by applying `sm.ols` function. The fitting is achieved by the attribute `fit()`. Results are stored in a structure that contains all necessary information. The attribute `params` extract a lot of data, such as the coefficients of the model and many others.

```
formula = 'y ~ x'
model = sm.ols(formula, df_data)
```

(Continued)